

Dynamic Macroeconomics II

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Fiscal Policies in the Growth Model

Third Part

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- The following equations are useful to understand effects:

$$c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} - g_t$$

$$q_t = \frac{\beta^t u'(c_t)}{1 + \tau_{ct}}$$

$$r_t = q_t f'(k_t)$$

$$w_t = q_t (f(k_t) - k_t f'(k_t))$$

$$u'(c_t) = \beta u'(c_{t+1}) R_{t+1}$$

$$R_{t+1} = \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} \left[\frac{1 - \tau_{it+1}}{1 - \tau_{it}} (1 - \delta) + \frac{1 - \tau_{kt+1}}{1 - \tau_{it}} f'(k_{t+1}) \right]$$

(As I said before, there should be a μ dividing in q_t)

Recall that we are working for now with the simplification that labor supply is inelastic. In this framework:

- A constant τ_c and a time-varying τ_n are not distorting.
- A constant tax on consumption disappears from gross real interest rate R_{t+1} . *No hay, ratio en el modelo, $U_{mgc}/U_{mgh} = W(1-\tau_n)$*
- The tax on labor income does not appear in system of equations.
- A time-varying consumption tax is distorting.
- Subsidy τ_i and tax τ_k always distort R_{t+1} , constant or time-varying.

check R_{t+1}

Experiments we can make

Once we have the model in the computer, there are two kinds of experiments we can make:

- **Impulse-response:** starting in a steady-state, change **one** exogenous variable for **one** period and observe response of the economy.
- Similar to taking a derivative.
- **Transition:** Take values of initial state variables from **data** in a given year, then introduce **observed** sequences of exogenous shocks.
- In this experiment there are two sources of dynamics: capital accumulation towards steady state + effect of exogenous variables.
- Application: analyze impact of fiscal reform Mexico 2014.

Implementing Shooting for economy with endogenous labor supply

Have one more unknown and one more equation. Assume no government for simplicity.

- $u(c_t, 1 - n_t) = \ln(c_t) + \psi \ln(1 - n_t)$.
- Feasibility: $c_t + k_{t+1} - (1 - \delta)k_t = y_t$.
- Production function: $y_t = f(k_t, n_t) = k_t^\alpha n_t^{1-\alpha}$.
- Euler equation (**check**):

$$\frac{c_{t+1}}{c_t} = \beta[\alpha k_{t+1}^{\alpha-1} n_{t+1}^{1-\alpha} + 1 - \delta]$$

- Consumption-leisure equation (**check**):

$$\frac{\psi c_t}{1 - n_t} = (1 - \alpha) k_t^\alpha n_t^{-\alpha}$$

- To solve an equilibrium with consumption-leisure decision, use the following algorithm:
 - 1 Starting with a given k_0 , make a guess for c_0 .
 - 2 **Extra step:** solve static consumption-leisure equation in $t = 0$ to get n_0 . A nonlinear equation, can use `fsolve` in Matlab.
 - 3 Use feasibility condition to obtain k_1 .
 - 4 Get c_1 from Euler equation.
 - 5 Repeat steps (2)-(4) to generate a sequence $\{k_0, k_1, \dots, k_S\}$.
 - 6 Compute the difference $k_S - \bar{k}$.
 - 7 If $k_S < \bar{k}$, reduce c_0 and recompute the sequence for capital.
 - 8 If $k_S > \bar{k}$, do the opposite.